The spherical average process of the whole-space LGF in odd dimensions

Abstract

For odd dimensions $d > 2$, we identify the spherical average process of the whole-space log-correlated Gaussian field in $\mathbb{R}^d$. More precisely, after parametrising the radius by e^{-t}, we show that this process is a causal Brownian moving average with kernel

$$q_d(u) = (d-2)e^{-2u}(1-e^{-2u})^{(d-4)/2}, \qquad u > 0.$$

The one-sided Fourier transform of this kernel is the gamma ratio

$$Q_d(\omega) = \frac{\Gamma(d/2)\Gamma(1+\mathrm{i}\omega/2)}{\Gamma(d/2+\mathrm{i}\omega/2)}.$$

For $d = 2m+1$, the corresponding power spectrum contains the non-rational factor $\omega\coth(\pi\omega/2)$. In contrast with the even-dimensional setting, the spherical average process therefore does not admit a finite-dimensional linear Gaussian Markov representation. We also establish the sharp regularity of the process and give an explicit infinite-dimensional Markovian representation. The construction of the odd-dimensional quantum cone is not considered here.

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1 Introduction

Let h be the whole-space log-correlated Gaussian field (LGF) on $\mathbb{R}^d$. This is the centred Gaussian generalised function defined modulo a global additive constant whose covariance kernel is $-\log|x-y|$. For $x \in \mathbb{R}^d$, the averages of h over the Euclidean spheres centred at x are well-defined modulo an additive constant. Upon parametrising the radius by e^{-t} and subtracting the average over the unit sphere, one obtains the centred Gaussian process

$$S_t \stackrel{\text{def}}{=} h_{e^{-t}}(x) - h_1(x), \qquad t \in \mathbb{R}. \tag{1}$$

In dimension $d = 2$, the process $(S_t)_{t\in\mathbb{R}}$ has the law of a standard two-sided Brownian motion.

The spherical average process in arbitrary dimension was introduced in the general framework of fractional Gaussian fields in [LSSW16, Section 11.1]; see also [DRSV17]. For even dimensions $d > 2$, Bertacco and Gwynne [BG25] identified $(S_t)_{t\in\mathbb{R}}$ with the integral of a finite-dimensional stationary Gaussian Markov process. This characterisation is one of the principal inputs in their construction of the higher-dimensional quantum cone. Their spherical-harmonic discussion also points to earlier coefficient-process computations of McKean [McK63]. Bertacco and Gwynne subsequently asked for an analogous characterisation in odd dimensions; see [BG25, Problem 2.16].

The purpose of this paper is to give such a characterisation. The main result shows that the Brownian moving-average formula appearing in the even-dimensional setting remains valid for every integer $d > 2$. The distinction between the two parities appears only after the formula has been obtained. If d is even, the moving-average kernel is a finite linear combination of exponential functions and hence gives a finite-dimensional Langevin equation. If d is odd, the exponent in the kernel is a half-integer, the corresponding expansion does not terminate, and the power spectrum is not rational.

The proof proceeds directly from the covariance formula for the spherical average process. We first take a mixed derivative in the logarithmic radii, which gives a stationary covariance kernel. We then compute its Fourier transform without imposing a parity assumption. The resulting power spectrum admits the explicit spectral factor

$$Q_d(\omega) = \frac{\Gamma(d/2)\Gamma(1+\mathrm{i}\omega/2)}{\Gamma(d/2+\mathrm{i}\omega/2)}. \tag{2}$$

A beta-integral computation shows that Q_d is the one-sided Fourier transform of the kernel q_d displayed in the abstract. This identifies the spherical average process as a Brownian moving average.

For $d = 2m+1$, Stirling's formula applied to (2) shows that the process is $(m-1)$ times continuously differentiable and that its $(m-1)$st derivative is locally α-Hölder continuous for every $\alpha \in (0,1)$. The next derivative does not exist in the mean-square sense. We also encode the nonlocal dependence on the past by a continuous memory profile indexed by $[0,\infty)$. This yields a time-homogeneous Markovian lift with state space $\mathbb{R} \times C_{\mathrm{loc}}([0,\infty))$.

We emphasise that the moving-average representation does not by itself construct the odd-dimensional quantum cone. Such a construction additionally requires a suitable mixing result for the recentered radial process. No such statement is proved in this paper.

2 Main results

Throughout the rest of the paper, let $d > 2$ be an integer and let h be a whole-space LGF on $\mathbb{R}^d$, defined modulo a global additive constant on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For $x \in \mathbb{R}^d$, let $(h_r(x))_{r>0}$ denote its spherical average process, and define $(S_t)_{t\in\mathbb{R}}$ by (1). We also let $(B_t)_{t\in\mathbb{R}}$ be a standard two-sided Brownian motion with $B_0 = 0$.

For $u > 0$, define

$$\begin{aligned} K_d(u) &\stackrel{\text{def}}{=} (1-e^{-2u})^{(d-2)/2}, \\ q_d(u) &\stackrel{\text{def}}{=} K_d'(u) = (d-2)e^{-2u}(1-e^{-2u})^{(d-4)/2}, \\ H_d(u) &\stackrel{\text{def}}{=} 1 - K_d(u). \end{aligned} \tag{3}$$

We extend K_d and H_d continuously to zero by setting $K_d(0) = 0$ and $H_d(0) = 1$.

2.1 Identification of the spherical average process

Theorem A (Identification of the spherical average process)**.** *For any $x \in \mathbb{R}^d$, the spherical average process $(S_t)_{t\in\mathbb{R}}$ admits a continuous modification. For this modification,*

$$(S_t)_{t\in\mathbb{R}} \overset{\text{law}}{=} \left(\int_0^\infty q_d(u)(B_{t-u} - B_{-u})\,\mathrm{d}u\right)_{t\in\mathbb{R}} \tag{4}$$

as random elements of $C_{\mathrm{loc}}(\mathbb{R})$. Equivalently,

$$(S_t)_{t\in\mathbb{R}} \overset{\text{law}}{=} \left((d-2)\int_{-\infty}^0 e^{2s}(1-e^{2s})^{(d-4)/2}(B_{s+t} - B_s)\,\mathrm{d}s\right)_{t\in\mathbb{R}}. \tag{5}$$

Moreover,

$$\int_0^\infty q_d(u)e^{-\mathrm{i}\omega u}\,\mathrm{d}u = Q_d(\omega) \overset{\text{def}}{=} \frac{\Gamma(d/2)\Gamma(1+\mathrm{i}\omega/2)}{\Gamma(d/2+\mathrm{i}\omega/2)}, \qquad \omega \in \mathbb{R}. \tag{6}$$

If $X = \partial_t S$ denotes the distributional derivative of S, then X is stationary and its covariance distribution R_d satisfies

$$\widehat{R}_d(\omega) = \frac{1}{\sqrt{2\pi}}|Q_d(\omega)|^2. \tag{7}$$

We observe that Theorem A is stated without a parity restriction. In even dimensions, (5) agrees with the representation in [BG25, Theorem A]. The new feature in odd dimensions is that this representation is no longer generated by finitely many ordinary derivatives.

2.2 Regularity in odd dimensions

Suppose now that

$$d = 2m+1, \qquad m \in \mathbb{N}. \tag{8}$$

In this case, the highest integer order of differentiability is

$$\mathfrak{c}_d \overset{\text{def}}{=} \frac{d-3}{2} = m-1. \tag{9}$$

For $k \in \mathbb{N}_0$ and $\alpha \in (0,1)$, we write $C^{k,\alpha}_{\mathrm{loc}}(\mathbb{R})$ for the space of functions which are locally k times continuously differentiable and whose kth derivative is locally α-Hölder continuous.

Theorem B (Regularity in odd dimensions)**.** *Let $d = 2m+1 \geq 3$. The process $(S_t)_{t\in\mathbb{R}}$ admits a modification such that*

$$S \in C^{m-1,\alpha}_{\mathrm{loc}}(\mathbb{R}), \qquad \forall \alpha \in (0,1). \tag{10}$$

Furthermore,

$$\mathbb{E}\left[\left(S^{(m-1)}_{t+r} - S^{(m-1)}_t\right)^2\right] \asymp r^2 \log\frac{1}{|r|}, \qquad r \to 0, \tag{11}$$

uniformly in $t \in \mathbb{R}$. An mth derivative does not exist in the mean-square sense.

The estimate (11) is the borderline case between Lipschitz regularity and the existence of one more derivative. In particular, the highest derivative in Theorem B is locally α-Hölder continuous for every $\alpha < 1$, but the spectral second moment required for an additional derivative diverges logarithmically.

2.3 The odd-dimensional spectrum

The gamma ratio in (6) has a particularly explicit form when d is odd.

Proposition 2.1 (Odd-dimensional power spectrum)**.** *Let $d = 2m + 1$. Then*

$$|Q_{2m+1}(\omega)|^2 = \frac{\pi((2m-1)!!)^2}{2} \frac{\omega \coth(\pi\omega/2)}{\prod_{j=0}^{m-1} (\omega^2 + (2j+1)^2)}, \qquad \omega \in \mathbb{R}, \tag{12}$$

where the value at $\omega = 0$ is defined by continuity and equals one. In particular, the meromorphic continuation of $|Q_{2m+1}|^2$ is not rational. Consequently, the stationary Gaussian process $X = \partial_t S$ is not the scalar output of any finite-dimensional, linear time-invariant Gaussian state-space model driven by white noise.

The final assertion is only an obstruction to a finite-dimensional linear Gaussian representation. It does not exclude an arbitrary nonlinear Markovian representation.

2.4 An infinite-dimensional Markovian representation

The nonlocal dependence on the past can be recorded explicitly. For $t \in \mathbb{R}$ and $r \geq 0$, define

$$M_t(r) \stackrel{\text{def}}{=} -\int_{-\infty}^{t} H_d(r + t - s)\, \mathrm{d}B_s. \tag{13}$$

Theorem C (The memory process)**.** *For every integer $d > 2$, the process in (13) admits a version such that $M_t \in C_{\mathrm{loc}}([0,\infty))$ for every $t \in \mathbb{R}$. With this choice of version, $(M_t)_{t\in\mathbb{R}}$ is a time-homogeneous Markov process on $C_{\mathrm{loc}}([0,\infty))$. More precisely, if*

$$W_v \stackrel{\text{def}}{=} B_{t+v} - B_t, \qquad v \geq 0,$$

then, for $u, r \geq 0$,

$$M_{t+u}(r) = M_t(r + u) - \int_0^u H_d(r + u - v)\, \mathrm{d}W_v, \tag{14}$$

and

$$S_{t+u} - S_t = M_t(u) - M_t(0) + \int_0^u K_d(u - v)\, \mathrm{d}W_v. \tag{15}$$

Consequently, the restriction $(S_t, M_t)_{t\geq 0}$ is a time-homogeneous Markovian lift of the spherical average process with state space $\mathbb{R} \times C_{\mathrm{loc}}([0,\infty))$. More generally, (14) and (15) define the corresponding Markovian evolution from any prescribed initial state.

2.5 Outline

In Section 3, we recall the definition of the whole-space LGF and its spherical average process, and record the covariance formula which will be used throughout the paper. In Section 4, we compute the power spectrum of the logarithmic radial derivative and prove Theorem A. In Section 5, we prove Theorems B and C and Theorem 2.1 and explain the distinction between the even- and odd-dimensional cases. Finally, in Section A, we record the specialisation to dimension three and the recovery of the even-dimensional Langevin equation.

3 Background and preliminaries

In this section, we recall the definition of the whole-space LGF and its spherical average process. We also record some elementary Fourier and gamma-function identities.

3.1 The whole-space log-correlated Gaussian field

For $d \geq 2$, let $\mathcal{S}(\mathbb{R}^d)$ be the Schwartz space and let

$$\mathcal{S}_0(\mathbb{R}^d) \stackrel{\text{def}}{=} \left\{ \varphi \in \mathcal{S}(\mathbb{R}^d) : \int_{\mathbb{R}^d} \varphi(x)\,\mathrm{d}x = 0 \right\}.$$

The whole-space LGF on $\mathbb{R}^d$ is the centred Gaussian random distribution h, defined modulo a global additive constant, whose covariance is given by

$$\mathbb{E}[(h,\varphi)(h,\psi)] = -\int_{\mathbb{R}^d}\int_{\mathbb{R}^d} \log|x-y|\varphi(x)\psi(y)\,\mathrm{d}x\,\mathrm{d}y, \qquad \varphi,\psi \in \mathcal{S}_0(\mathbb{R}^d). \tag{16}$$

Equivalently, h is the standard Gaussian on the homogeneous Sobolev space of order $d/2$; see [DRSV17, LSSW16].

The law of h modulo constants is invariant under translations, rotations and dilations. In particular, the logarithmic radial process in (1) has stationary increments.

3.2 The spherical average

For a continuous function $f : \mathbb{R}^d \to \mathbb{R}$, its spherical average around $x \in \mathbb{R}^d$ is

$$f_r(x) \stackrel{\text{def}}{=} \frac{1}{|\mathbb{S}^{d-1}|}\int_{\mathbb{S}^{d-1}} f(x + r\vartheta)\,\mathrm{d}\vartheta, \qquad r > 0.$$

For the LGF, the same object is defined in the sense of distributions. More precisely, for $\phi \in C_c^\infty(\mathbb{R}_+) \cap \mathcal{S}_0(\mathbb{R})$, set

$$(h_\cdot(x),\phi) \stackrel{\text{def}}{=} \frac{1}{|\mathbb{S}^{d-1}|}\left(h, z \mapsto \frac{\phi(|z-x|)}{|z-x|^{d-1}}\right). \tag{17}$$

This definition agrees with the usual spherical average when h is a continuous function; see [LSSW16, Section 11.1] and [BG25, Section 3.2].

Averaging the covariance kernel $-\log|x-y|$ over two spheres centred at the same point gives the following covariance representative:

$$\mathcal{K}_d(r_1,r_2) = -c(d)\int_0^\pi \log(r_1^2 + r_2^2 - 2r_1r_2\cos\vartheta)(\sin\vartheta)^{d-2}\,\mathrm{d}\vartheta, \tag{18}$$

where

$$c(d) \stackrel{\text{def}}{=} \frac{\Gamma(d/2)}{2\sqrt{\pi}\,\Gamma((d-1)/2)}. \tag{19}$$

Since the field is defined modulo constants, a different covariance representative can differ from (18) by a function of r_1 plus a function of r_2. This ambiguity disappears after taking one logarithmic derivative in each variable.

3.3 Basic and recurrent notation

We write $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$ and $\mathbb{R}_+ = (0,\infty)$. For non-negative quantities a and b, we write $a \lesssim b$ if there exists a constant $C > 0$, independent of the parameters of interest, such that $a \leq Cb$. We write $a \asymp b$ if $a \lesssim b$ and $b \lesssim a$.

We use the unitary Fourier transform

$$\widehat{f}(\omega) = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} f(t)e^{-\mathrm{i}\omega t}\,\mathrm{d}t, \qquad f(t) = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} \widehat{f}(\omega)e^{\mathrm{i}\omega t}\,\mathrm{d}\omega. \tag{20}$$

We will repeatedly use

$$|\Gamma(1+\mathrm{i}x)|^2 = \frac{\pi x}{\sinh(\pi x)}, \qquad |\Gamma(1/2+\mathrm{i}x)|^2 = \frac{\pi}{\cosh(\pi x)}, \tag{21}$$

as well as the duplication formula

$$\Gamma(2z) = \frac{2^{2z-1}}{\sqrt{\pi}}\Gamma(z)\Gamma(z+1/2). \tag{22}$$

4 Identification of the spherical average process

The aim of this section is to prove Theorem A. We first compute the covariance of the distributional derivative of the spherical average process, then evaluate its Fourier transform, and finally identify a causal spectral factor.

4.1 The covariance of the radial derivative

Let $X = \partial_t S$ in the sense of random distributions.

Proposition 4.1 (Covariance of the radial derivative)**.** *The process X is a centred stationary Gaussian random distribution. Its covariance is represented by the locally integrable function*

$$R_d(u) = c(d) \int_0^\pi \frac{1 - \cos\vartheta \cosh u}{(\cosh u - \cos\vartheta)^2} (\sin\vartheta)^{d-2} \, \mathrm{d}\vartheta, \qquad u \neq 0. \tag{23}$$

Moreover, $R_d \in L^1(\mathbb{R})$.

Proof. Substitute $r_1 = e^{-t}$ and $r_2 = e^{-s}$ in (18). Since

$$e^{-2t} + e^{-2s} - 2e^{-t-s}\cos\vartheta = 2e^{-(t+s)}(\cosh(t-s) - \cos\vartheta),$$

the mixed derivative in t and s only depends on $u = t - s$. A direct differentiation gives (23) away from $u = 0$.

We next check that no Dirac mass appears at the origin. Let

$$F(u) \overset{\text{def}}{=} \int_0^\pi \log(\cosh u - \cos\vartheta)(\sin\vartheta)^{d-2} \, \mathrm{d}\vartheta.$$

For $|u| \leq 1$, splitting the integral at $\vartheta = 1$ gives

$$|F'(u)| \lesssim |u| \int_0^1 \frac{\vartheta^{d-2}}{u^2 + \vartheta^2} \, \mathrm{d}\vartheta + |u|.$$

The right-hand side is $O(|u|\log(1/|u|))$ when $d = 3$ and $O(|u|)$ when $d > 3$. In particular, F' is continuous at zero and therefore its distributional derivative has no atomic part.

Finally, in a neighbourhood of $(u, \vartheta) = (0, 0)$, the absolute value of the integrand in (23) is bounded by

$$C \frac{\vartheta^{d-2}}{u^2 + \vartheta^2}.$$

This is jointly integrable for $d > 2$. For large $|u|$, the integrand decays exponentially. Hence $R_d \in L^1(\mathbb{R})$. □

4.2 The power spectrum

We next compute the Fourier transform of (23). The first lemma is the one-dimensional transform which also appears in the even-dimensional calculation of [BG25, Section 4].

Lemma 4.2 (A hyperbolic Fourier transform)**.** *For $\vartheta \in (0, \pi)$, let*

$$\psi_\vartheta(u) \overset{\text{def}}{=} \frac{1 - \cos\vartheta\cosh u}{(\cosh u - \cos\vartheta)^2}, \qquad u \in \mathbb{R}.$$

Then

$$\widehat{\psi_\vartheta}(\omega) = \frac{1}{\sqrt{2\pi}} \frac{2\pi\omega\cosh((\pi - \vartheta)\omega)}{\sinh(\pi\omega)}, \qquad \omega \in \mathbb{R}, \tag{24}$$

where the value at $\omega = 0$ is defined by continuity.

Proof. The standard strip Poisson-kernel identity is

$$\int_{\mathbb{R}} \frac{\sin\vartheta}{\cosh u - \cos\vartheta} e^{-\mathrm{i}\omega u}\,\mathrm{d}u = \frac{2\pi\sinh((\pi-\vartheta)\omega)}{\sinh(\pi\omega)}. \tag{25}$$

It follows, for example, by integrating over a rectangular contour of height $2\pi\mathrm{i}$ and summing the residues at the two poles in the strip. Since

$$\psi_\vartheta(u) = -\partial_\vartheta\left(\frac{\sin\vartheta}{\cosh u - \cos\vartheta}\right),$$

differentiating (25) with respect to ϑ and using the normalisation in (20) gives (24). □

Lemma 4.3 (An angular beta integral)**.** *For every integer $d \geq 2$ and $\omega \in \mathbb{R}$,*

$$\int_0^\pi e^{\omega(\vartheta-\pi/2)}(\sin\vartheta)^{d-2}\,\mathrm{d}\vartheta = \frac{\pi\Gamma(d-1)}{2^{d-2}\Gamma(d/2+\mathrm{i}\omega/2)\Gamma(d/2-\mathrm{i}\omega/2)}. \tag{26}$$

Proof. For $n \in \mathbb{N}_0$, set

$$J_n(\omega) \stackrel{\text{def}}{=} \int_0^\pi e^{\omega(\vartheta-\pi/2)}\sin^n\vartheta\,\mathrm{d}\vartheta.$$

Direct integration gives

$$J_0(\omega) = \frac{2\sinh(\pi\omega/2)}{\omega}, \qquad J_1(\omega) = \frac{2\cosh(\pi\omega/2)}{1+\omega^2},$$

with the values at zero defined by continuity. For $n \geq 2$, integration by parts gives

$$J_n(\omega) = \frac{n(n-1)}{n^2+\omega^2}J_{n-2}(\omega). \tag{27}$$

The right-hand side of (26), with $d = n+2$, has the same two initial values and satisfies the same recurrence by $\Gamma(z+1) = z\Gamma(z)$. This proves the claim. □

Proposition 4.4 (Power spectrum)**.** *For every integer $d > 2$, the Fourier transform of the covariance kernel in (23) is*

$$\widehat{R}_d(\omega) = \frac{1}{\sqrt{2\pi}}\left|\frac{\Gamma(d/2)\Gamma(1+\mathrm{i}\omega/2)}{\Gamma(d/2+\mathrm{i}\omega/2)}\right|^2. \tag{28}$$

Proof. The estimates in the proof of Theorem 4.1 justify Fubini's theorem. Hence, by Theorem 4.2,

$$\widehat{R}_d(\omega) = \frac{1}{\sqrt{2\pi}}\frac{2\pi c(d)\omega}{\sinh(\pi\omega)}\int_0^\pi \cosh((\pi-\vartheta)\omega)(\sin\vartheta)^{d-2}\,\mathrm{d}\vartheta. \tag{29}$$

By symmetry about $\pi/2$ and Theorem 4.3,

$$\int_0^\pi \cosh((\pi-\vartheta)\omega)(\sin\vartheta)^{d-2}\,\mathrm{d}\vartheta = \frac{\pi\Gamma(d-1)\cosh(\pi\omega/2)}{2^{d-2}|\Gamma(d/2+\mathrm{i}\omega/2)|^2}. \tag{30}$$

Combining (29) and (30), and then applying (21), (22), and $\sinh(\pi\omega) = 2\sinh(\pi\omega/2)\cosh(\pi\omega/2)$, gives (28). □

4.3 Proof of the moving-average representation

We are now ready to prove Theorem A.

Proof of Theorem A. The kernel q_d in (3) satisfies

$$q_d(u) \sim (d-2)(2u)^{(d-4)/2} \quad (u \downarrow 0), \qquad q_d(u) = O(e^{-2u}) \quad (u \to \infty). \tag{31}$$

In particular, $q_d \in L^1(\mathbb{R}_+)$ and

$$\int_0^\infty q_d(u)\,\mathrm{d}u = 1. \tag{32}$$

Almost surely, a two-sided Brownian path has at most polynomial growth. Consequently, the integral on the right-hand side of (4) is an ordinary absolutely convergent Lebesgue integral for every $t \in \mathbb{R}$.

Let

$$T_t \stackrel{\text{def}}{=} \int_0^\infty q_d(u)(B_{t-u} - B_{-u})\,\mathrm{d}u.$$

For fixed t, the integral also converges in L^2. Indeed, the covariance of two Brownian increments of length $|t|$ has absolute value at most $|t|$, so

$$\mathbb{E}[T_t^2] \le |t| \left(\int_0^\infty q_d(u)\,\mathrm{d}u \right)^2 = |t|.$$

Approximating the integral by finite sums shows that T is a centred Gaussian process.

We next establish continuity. Fix $A > 0$ and split the integral at $U > A + 1$. On $u \le U$, continuity is uniform for $|t| \le A$ by uniform continuity of the Brownian path on compact intervals. On $u \ge U$, the polynomial growth of the Brownian path and the exponential decay in (31) give an integrable path-dependent majorant, uniform for $|t| \le A$. Dominated convergence therefore gives continuity on $[-A, A]$. Since A is arbitrary, T is almost surely continuous.

For the one-sided Fourier transform, the change of variables $y = e^{-2u}$ gives

$$\begin{aligned}
\int_0^\infty q_d(u) e^{-\mathrm{i}\omega u}\,\mathrm{d}u &= \frac{d-2}{2} \int_0^1 y^{\mathrm{i}\omega/2} (1-y)^{(d-4)/2}\,\mathrm{d}y \\
&= \frac{d-2}{2} B\left(1 + \frac{\mathrm{i}\omega}{2}, \frac{d-2}{2}\right) \\
&= \frac{\Gamma(d/2)\Gamma(1 + \mathrm{i}\omega/2)}{\Gamma(d/2 + \mathrm{i}\omega/2)}.
\end{aligned} \tag{33}$$

In the sense of random tempered distributions, $\partial_t T = q_d * \dot{B}$. Hence the Fourier transform of the covariance distribution of $\partial_t T$ is $|Q_d|^2/\sqrt{2\pi}$. By Theorem 4.4, the same is true for $\partial_t S$.

It remains to recover the additive constants. Both S and T vanish at zero. Let

$$D(t,s) \stackrel{\text{def}}{=} \mathbb{E}[S_t S_s] - \mathbb{E}[T_t T_s].$$

The covariance of S is continuous by the local integrability of the logarithmic singularity in (18), and the covariance of T is continuous by the construction above. Since $\partial_t \partial_s D = 0$ distributionally, there exist continuous functions a and b such that $D(t,s) = a(t) + b(s)$. The identities $D(t,0) = D(0,s) = 0$ imply that $D \equiv 0$. Thus S and T have the same covariance and therefore the same law. The representation (5) follows by the change of variables $s = -u$. □

We conclude this section with a representation which will be used below.

Corollary 4.5 (Brownian motion plus a stationary process)**.** *Let*

$$Z_t \stackrel{\text{def}}{=} -\int_{-\infty}^t H_d(t-s)\,\mathrm{d}B_s. \tag{34}$$

Then Z is a stationary continuous Gaussian process and

$$S_t = B_t + Z_t - Z_0, \qquad t \in \mathbb{R}, \tag{35}$$

almost surely for the moving-average coupling of Theorem A. In particular,

$$\mathbb{E}[S_t^2] = |t| + O(1), \qquad |t| \to \infty. \tag{36}$$

Proof. Since $H_d(0) = 1$ and $H_d(u) = O(e^{-2u})$ as $u \to \infty$, one has $H_d \in L^1(\mathbb{R}_+) \cap L^2(\mathbb{R}_+)$. Stochastic integration by parts in (4), first on a finite interval and then in L^2, gives (35) for each fixed t.

The process Z is stationary. Moreover, for $0 < r \le 1$, Itô's isometry gives

$$\mathbb{E}[(Z_{t+r} - Z_t)^2] = \int_0^\infty |H_d(v+r) - H_d(v)|^2 \, \mathrm{d}v + \int_0^r H_d(v)^2 \, \mathrm{d}v \lesssim r.$$

Gaussian moment estimates and the Kolmogorov–Chentsov theorem give a continuous modification. Therefore (35) holds simultaneously for all t after choosing continuous versions.

For $t > 0$, $\mathrm{Cov}(B_t, Z_0) = 0$ and

$$\mathrm{Cov}(B_t, Z_t) = -\int_0^t H_d(u) \, \mathrm{d}u,$$

which is uniformly bounded in t. Also, $\mathbb{E}[(Z_t - Z_0)^2]$ is uniformly bounded. This gives (36) for $t > 0$, and the case $t < 0$ follows from stationary increments. □

5 Odd-dimensional regularity and Markovian structure

Throughout this section, unless stated otherwise, we assume that $d = 2m + 1 \ge 3$.

5.1 The odd-dimensional power spectrum

We first prove Theorem 2.1.

Proof of Theorem 2.1. Set $x = \omega/2$. By (21) and the recurrence relation for the gamma function,

$$\begin{aligned} |Q_{2m+1}(\omega)|^2 &= \Gamma(m+1/2)^2 \frac{|\Gamma(1+\mathrm{i}x)|^2}{|\Gamma(m+1/2+\mathrm{i}x)|^2} \\ &= \Gamma(m+1/2)^2 \frac{x \coth(\pi x)}{\prod_{j=0}^{m-1} (x^2 + (j+1/2)^2)}. \end{aligned}$$

Using

$$\Gamma(m+1/2) = \frac{(2m-1)!!}{2^m}\sqrt{\pi}$$

and substituting $x = \omega/2$ gives (12).

The right-hand side of (12), viewed as a meromorphic function of a complex variable z, has poles at $z = 2\mathrm{i}k$ for infinitely many non-zero integers k. The finitely many zeros of the denominator at odd imaginary integers are cancelled by zeros of $\coth(\pi z/2)$. Hence the meromorphic continuation is not rational.

For a finite-dimensional stable linear system

$$\mathrm{d}Y_t = AY_t \, \mathrm{d}t + b \, \mathrm{d}W_t, \qquad X_t = c^\top Y_t,$$

the transfer function $c^\top(\mathrm{i}\omega I - A)^{-1}b$ is rational, and therefore the power spectrum of X is rational; see, e.g., [PP02, Section 10.3]. This proves the final assertion. □

For the first three odd dimensions, the kernels and spectra are

5.2 Proof of the regularity theorem

Stirling's formula applied to (6) gives

$$|Q_d(\omega)|^2 \sim 2^{d-2}\Gamma(d/2)^2 |\omega|^{2-d}, \qquad |\omega| \to \infty. \tag{37}$$

d	$q_d(u)$	$\lvert Q_d(\omega)\rvert^2$
3	$\dfrac{e^{-2u}}{\sqrt{1-e^{-2u}}}$	$\dfrac{\pi\omega}{2(1+\omega^2)}\coth\left(\dfrac{\pi\omega}{2}\right)$
5	$3e^{-2u}\sqrt{1-e^{-2u}}$	$\dfrac{9\pi\omega}{2(1+\omega^2)(9+\omega^2)}\coth\left(\dfrac{\pi\omega}{2}\right)$
7	$5e^{-2u}(1-e^{-2u})^{3/2}$	$\dfrac{225\pi\omega}{2(1+\omega^2)(9+\omega^2)(25+\omega^2)}\coth\left(\dfrac{\pi\omega}{2}\right)$

Table 1: The first three odd-dimensional kernels and power spectra.

Proof of Theorem B. The spectral measure of $X = \partial_t S$ is

$$\frac{1}{2\pi}|Q_{2m+1}(\omega)|^2\,\mathrm{d}\omega.$$

For $1 \le k \le m-1$, the kth mean-square derivative of S exists since

$$\int_{\mathbb{R}} |\omega|^{2(k-1)}|Q_{2m+1}(\omega)|^2\,\mathrm{d}\omega < \infty.$$

Indeed, by (37), the integrand has order $|\omega|^{2k-2m-1}$ at infinity, which is integrable precisely when $k < m$.

Let $Y = S^{(m-1)}$, with the convention $Y = S$ when $m = 1$. Set

$$\rho_m(\omega) \overset{\text{def}}{=} \begin{cases} |\omega|^{-2}|Q_3(\omega)|^2, & m = 1, \\ |\omega|^{2m-4}|Q_{2m+1}(\omega)|^2, & m \ge 2. \end{cases}$$

For $m \ge 2$, $\rho_m/(2\pi)$ is the spectral density of the stationary process Y. For $m = 1$, it is the spectral weight governing the stationary increments of S. In both cases,

$$\rho_m(\omega) \asymp |\omega|^{-3}, \qquad |\omega| \to \infty. \tag{38}$$

Consequently,

$$\mathbb{E}[(Y_{t+r} - Y_t)^2] = \frac{1}{\pi}\int_{\mathbb{R}} (1-\cos(r\omega))\rho_m(\omega)\,\mathrm{d}\omega. \tag{39}$$

The contribution from $|\omega| \le 1$ is $O(r^2)$. On $1 \le |\omega| \le |r|^{-1}$, one has $1 - \cos(r\omega) \asymp r^2\omega^2$, so (38) gives a contribution comparable to

$$r^2\int_1^{|r|^{-1}} \frac{\mathrm{d}\omega}{\omega} = r^2 \log\frac{1}{|r|}.$$

The contribution from $|\omega| \ge |r|^{-1}$ is $O(r^2)$. This proves (11).

All moments of a centred Gaussian increment are controlled by its variance. Fix $\alpha \in (0,1)$ and choose $\beta \in (\alpha, 1)$. Since

$$r^2\log(1/|r|) \lesssim |r|^{2\beta},$$

Gaussian moment estimates and the Kolmogorov–Chentsov theorem give a locally α-Hölder continuous modification of Y. Taking a countable intersection over rational $\alpha \in (0,1)$ gives one modification which has this property for every $\alpha < 1$.

For $m \ge 2$, the lower derivatives are recovered recursively by pathwise integration. The corresponding identities hold first in L^2 by spectral approximation and then simultaneously for all times by continuity. This gives (10).

Finally, an mth mean-square derivative would require

$$\int_{\mathbb{R}} \omega^2\rho_m(\omega)\,\mathrm{d}\omega < \infty,$$

which fails logarithmically by (38). □

5.3 The distinction between even and odd dimensions

The parity distinction is already visible in the binomial expansion of q_d. If $d = 2n$ is even, then

$$q_{2n}(u) = (2n-2)\sum_{j=0}^{n-2}(-1)^j\binom{n-2}{j}e^{-2(j+1)u}. \tag{40}$$

Thus q_{2n} is a finite linear combination of exponential kernels. This yields the finite-dimensional Langevin equation of [BG25, Theorem A].

If $d = 2m+1$ is odd, then

$$q_{2m+1}(u) = (2m-1)\sum_{j=0}^{\infty}(-1)^j\binom{m-3/2}{j}e^{-2(j+1)u}, \qquad u > 0. \tag{41}$$

The expansion does not terminate. Equivalently, the factor $\omega\coth(\pi\omega/2)$ in (12) prevents the power spectrum from being rational.

The gamma ratio also gives a nonlocal Langevin equation. Define the Fourier multiplier $\mathcal{L}_d$ on stationary generalised processes by

$$\widehat{\mathcal{L}_d f}(\omega) \stackrel{\text{def}}{=} \frac{\Gamma(d/2 + \mathrm{i}\omega/2)}{\Gamma(d/2)\Gamma(1+\mathrm{i}\omega/2)}\widehat{f}(\omega). \tag{42}$$

For the moving-average coupling of Theorem A,

$$\mathcal{L}_d X = \dot{B} \tag{43}$$

as an equality of stationary random tempered distributions. If $d = 2n$, the multiplier in (42) is a polynomial in $\mathrm{i}\omega$. If d is odd, it is non-polynomial and (43) is genuinely nonlocal.

5.4 Proof of the memory representation

We now prove Theorem C.

Proof of Theorem C. The stochastic integral in (13) is well-defined since $H_d \in L^2(\mathbb{R}_+)$. For $r, r' \geq 0$, Itô's isometry gives

$$\mathbb{E}[|M_t(r') - M_t(r)|^2] = \int_0^\infty |H_d(r'+v) - H_d(r+v)|^2\,\mathrm{d}v. \tag{44}$$

If $d \geq 4$, then $H_d' = -q_d \in L^2(\mathbb{R}_+)$ and the right-hand side of (44) is bounded by

$$|r'-r|^2\|q_d\|^2_{L^2(\mathbb{R}_+)}.$$

If $d = 3$, suppose without loss of generality that $r' = r + \delta$ with $0 < \delta \leq 1/2$. The estimate

$$|K_3(x) - K_3(y)| \lesssim |x-y|^{1/2}, \qquad x, y \in [0,2],$$

controls the part of the integral with $0 \leq v \leq \delta$. On $\delta \leq v \leq 1$, the mean-value theorem and $q_3(v) \lesssim v^{-1/2}$ give

$$|H_3(v+\delta) - H_3(v)|^2 \lesssim \frac{\delta^2}{v}.$$

The remaining part is $O(\delta^2)$ by exponential decay. Hence, locally uniformly in r and r',

$$\mathbb{E}[|M_t(r') - M_t(r)|^2] \lesssim |r'-r|^2\log\frac{e}{|r'-r|}. \tag{45}$$

Gaussian moment estimates and the Kolmogorov–Chentsov theorem therefore give a continuous modification of $r \mapsto M_t(r)$ on every compact interval. The same computation in the time variable gives, locally uniformly in r,

$$\mathbb{E}[|M_{t+u}(r) - M_t(r)|^2] \lesssim |u|.$$

By Gaussian moment estimates and the multiparameter Kolmogorov–Chentsov theorem, M therefore admits a version which is jointly continuous in (t, r) on compact subsets of $\mathbb{R} \times [0, \infty)$. In particular, $M_t \in C_{\mathrm{loc}}([0, \infty))$ for every $t \in \mathbb{R}$.

Fix $t \in \mathbb{R}$ and set $W_v = B_{t+v} - B_t$. Splitting the stochastic integral at time t gives, for $u, r \geq 0$,

$$M_{t+u}(r) = M_t(r+u) - \int_0^u H_d(r+u-v)\,\mathrm{d}W_v,$$

which is (14). Furthermore, $M_t(0) = Z_t$, where Z is defined in (34). Combining (35) with (14) at $r = 0$ gives

$$\begin{aligned} S_{t+u} - S_t &= W_u + M_{t+u}(0) - M_t(0) \\ &= M_t(u) - M_t(0) + \int_0^u (1 - H_d(u-v))\,\mathrm{d}W_v, \end{aligned}$$

which is (15) since $K_d = 1 - H_d$.

Let $\mathcal{G}_t$ be the sigma-field generated by the driving white noise on $(-\infty, t]$. The profile M_t is $\mathcal{G}_t$-measurable, while $(W_v)_{v \geq 0}$ is independent of $\mathcal{G}_t$. Equation (14) therefore proves that M is a time-homogeneous Markov process. Moreover, for $t \geq 0$, the variable S_t is $\mathcal{G}_t$-measurable by (35). Hence (14) and (15) show that $(S_t, M_t)_{t \geq 0}$ is a time-homogeneous Markov process. The same equations define the transition from an arbitrary initial state. □

Remark 5.1. The memory state in Theorem C is not claimed to be minimal, nor is it claimed to be measurable with respect to the natural filtration generated by S alone. Its role is to give an explicit Markovian realisation of the moving-average process.

A Consistency checks

A.1 Dimension three

For $d = 3$,

$$q_3(u) = \frac{e^{-2u}}{\sqrt{1 - e^{-2u}}}, \qquad Q_3(\omega) = \frac{\sqrt{\pi}}{2} \frac{\Gamma(1 + \mathrm{i}\omega/2)}{\Gamma(3/2 + \mathrm{i}\omega/2)}. \tag{46}$$

Consequently,

$$|Q_3(\omega)|^2 = \frac{\pi\omega}{2(1+\omega^2)} \coth\left(\frac{\pi\omega}{2}\right). \tag{47}$$

The kernel has the integrable singularity $q_3(u) \sim (2u)^{-1/2}$ at zero. The process satisfies

$$\mathbb{E}[(S_{t+r} - S_t)^2] \asymp r^2 \log \frac{1}{|r|}, \qquad r \to 0,$$

so it is locally α-Hölder continuous for every $\alpha < 1$ and has no mean-square derivative.

A.2 Recovery of the even-dimensional equation

Let $d = 2n$. The gamma recurrence gives

$$Q_{2n}(\omega) = \frac{2^{n-1}\Gamma(n)}{\prod_{j=1}^{n-1}(2j + \mathrm{i}\omega)}, \tag{48}$$

and hence

$$|Q_{2n}(\omega)|^2 = \frac{2^{2n-2}\Gamma(n)^2}{\prod_{j=1}^{n-1}(\omega^2 + 4j^2)}. \tag{49}$$

Equivalently, if $X = \partial_t S$, then

$$\prod_{j=1}^{n-1} (\partial_t + 2j) X = 2^{n-1} \Gamma(n) \dot{B} \tag{50}$$

as stationary random distributions. This is the scalar higher-order form of the companion-matrix Langevin equation in [BG25, Theorem A].

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